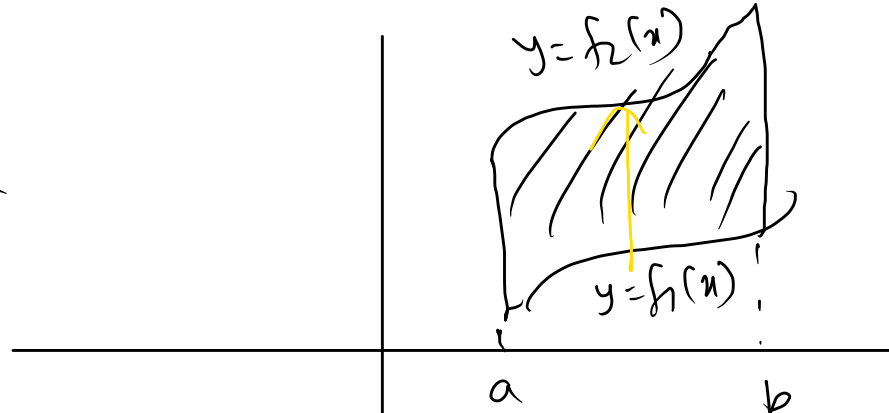
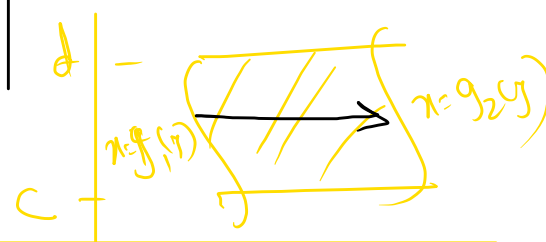


$$\int_{x=a}^{x=b} \int_{y=f_1(x)}^{y=f_2(x)} f(x,y) \, dy \, dx$$

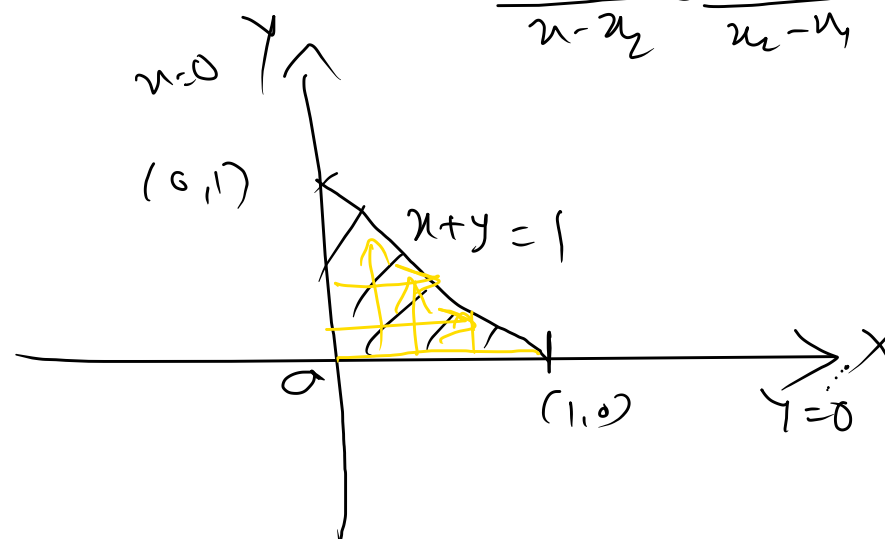


$$\int_{y=c}^{y=d} \int_{x=g_1(y)}^{x=g_2(y)} f(x,y) \, dx \, dy$$



$$\frac{y-y_2}{x-x_2} = \frac{y_2-y_1}{x_2-x_1}$$

$$\int_{x=0}^1 \int_{y=0}^{1-x} f(x,y) \, dy \, dx$$



~~$$\int_0^1 \int_0^{1-x} f \, dy \, dx = \int_0^1 (y) \Big|_0^{1-x} dx = \int_0^1 (1-x) dx$$~~

$$= \left(x - \frac{x^2}{2} \right) \Big|_0^1$$

$\iint xy \, dx \, dy$ over the region bounded by

x -axis, ordinate at $x = 2a$, parabola $x^2 = 4ay$

$$x^2 = 4ay$$

$$x = \pm \sqrt{4ay}$$

$$x = 2a \quad \& \quad x^2 = 4ay$$

$$4a^2 = 4ay$$

$$\int_{x=0}^{2a} \int_{y=0}^{x^2/4a} xy \, dy \, dx$$

$$= \int_0^{2a} x \left(\frac{y^2}{2} \right)_0^{x^2/4a} dx$$

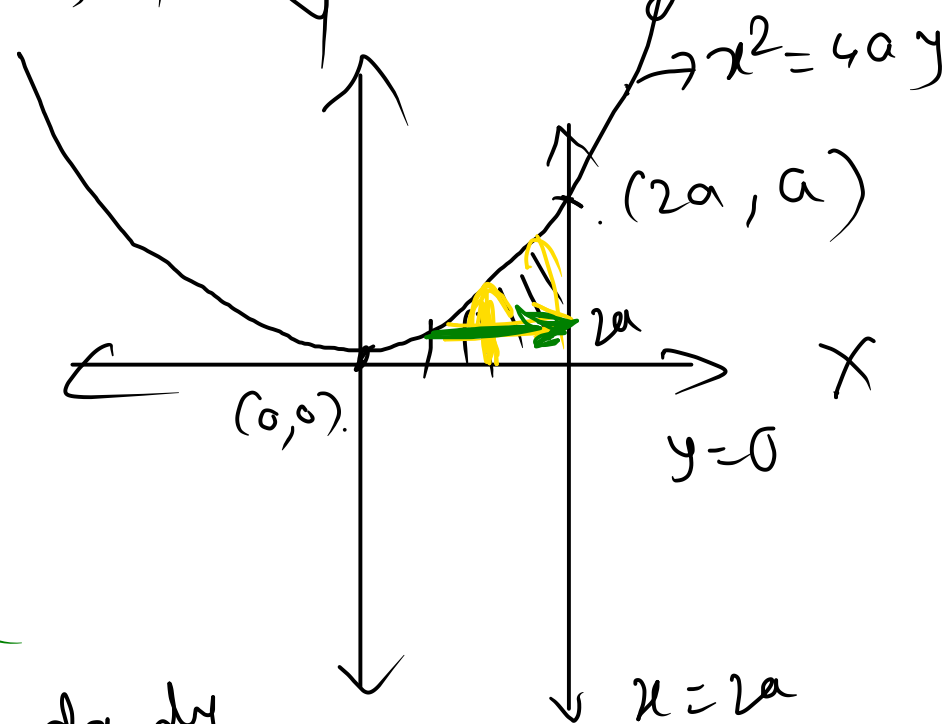
$$\int_{y=0}^a \int_{x=\sqrt{4ay}}^{2a} xy \, dx \, dy$$

$$= \int_0^a x \left(\frac{x^4}{150^2 \times 2} \right) dx = \frac{a^4}{3}$$

$$x^2 = 4a$$

$$= \frac{1}{32a^2} \int_0^{2a} x^5 dx$$

$$= \frac{1}{32a^2} \left(\frac{x^6}{6} \right)_0^{2a} = \frac{1}{32a^2} \times \frac{64a^6}{6} = \frac{a^4}{3}$$

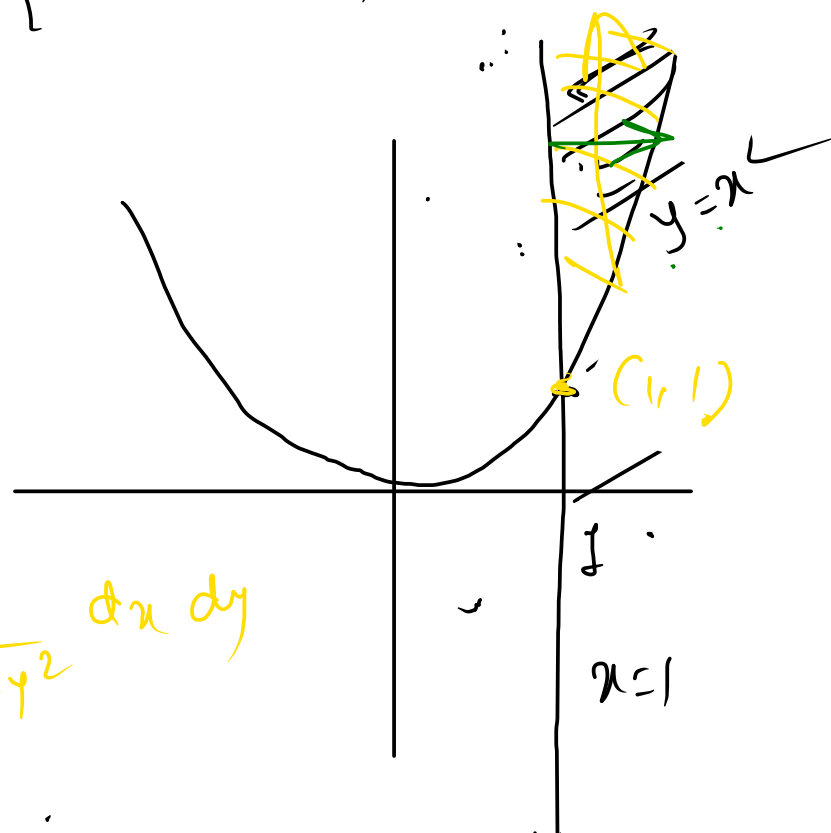


$$\iint_R \frac{1}{x^2+y^2} dx dy \quad \text{where } R \text{ is } x \geq 1, y \geq x^2.$$

$$x=1$$

$$y=x^2$$

$$x = \pm \sqrt{y}$$



$$\int_{x=1}^{\infty} \int_{y=x^2}^{\infty} \frac{1}{x^4+y^2} dy dx$$

$$\int_{y=1}^{\infty} \int_{x=1}^{\sqrt{y}} \frac{1}{x^4+y^2} dx dy$$

$$\int_1^{\infty} \int_{x^2}^{\infty} \frac{1}{y^2+(x^2)^2} dy dx$$

$$= \int_1^{\infty} \left[\frac{1}{x^2} \tan^{-1} \left(\frac{y}{x^2} \right) \right]_{x^2}^{\infty} dx$$

$$= \int_1^{\infty} \left[\frac{1}{x^2} \left(\frac{\pi}{2} \right) - \frac{1}{x^2} \frac{\pi}{4} \right] dx$$

$$= \int_1^{\infty} \left(\frac{\pi}{2} - \frac{\pi}{4} \right) x^{-2} dx$$

$$= \left(\frac{\pi}{4} \right) \left(\frac{x^{-1}}{-1} \right)_1^{\infty} = \frac{\pi}{4} \left(-\frac{1}{x} \right)_1^{\infty}$$

$$= \frac{\pi}{4} [0 + 1]$$

$\iint xy \, dx \, dy$ over area bdd by parabolas
 $y = x^2$ & $x = -y^2$



$$y = x^2$$

$$\Rightarrow x = \pm \sqrt{y}$$

$$y = x^2$$

$$x = -y^2$$

$$y = y^4$$

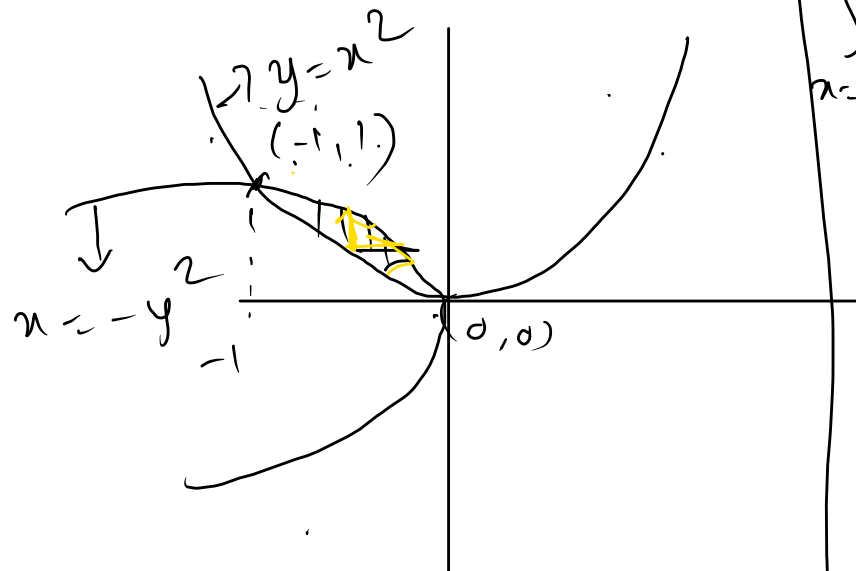
$$x = -y^2$$

$$y^2 = -x$$

$$y = \pm \sqrt{-x}$$

$$y(y^3 - 1) = 0 \Rightarrow y = 0 \text{ or } y = 1$$

for $y = 0$ as $y = x^2 \Rightarrow x = 0$
 $y = 1$ as $1 = x^2 \Rightarrow x = \pm 1$



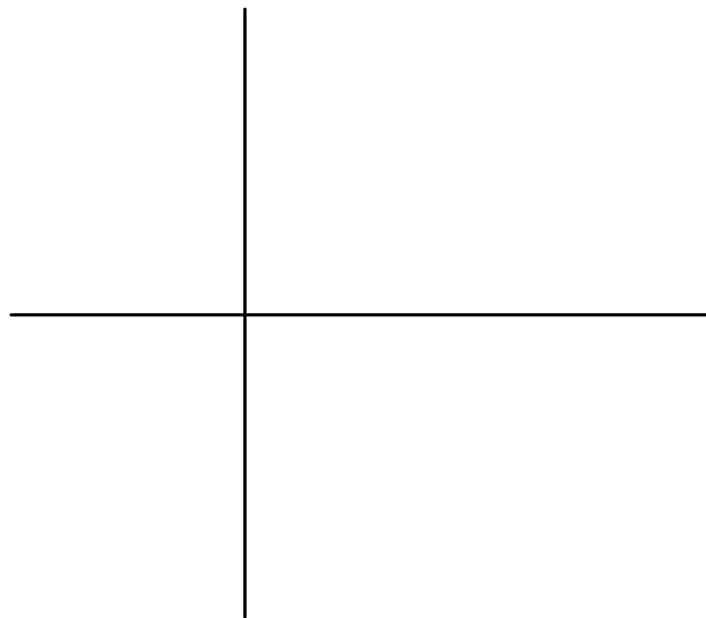
$$\int_0^1 \int_{-y^2}^{\sqrt{y}} xy \, dx \, dy$$

$$= -\frac{1}{12}$$

$$\int_0^1 \int_{-y^2}^{\sqrt{y}} xy \, dx \, dy$$

$$y = 0 \text{ or } x = -\sqrt{y}$$

$\iint xy \, dx \, dy$ over area bdd by parabolas
 $y = x^2$ & $x = -y^2$



1



dy



$\iint y \, dx \, dy$ over the area bdd by $x=0$, $y=x^2$,

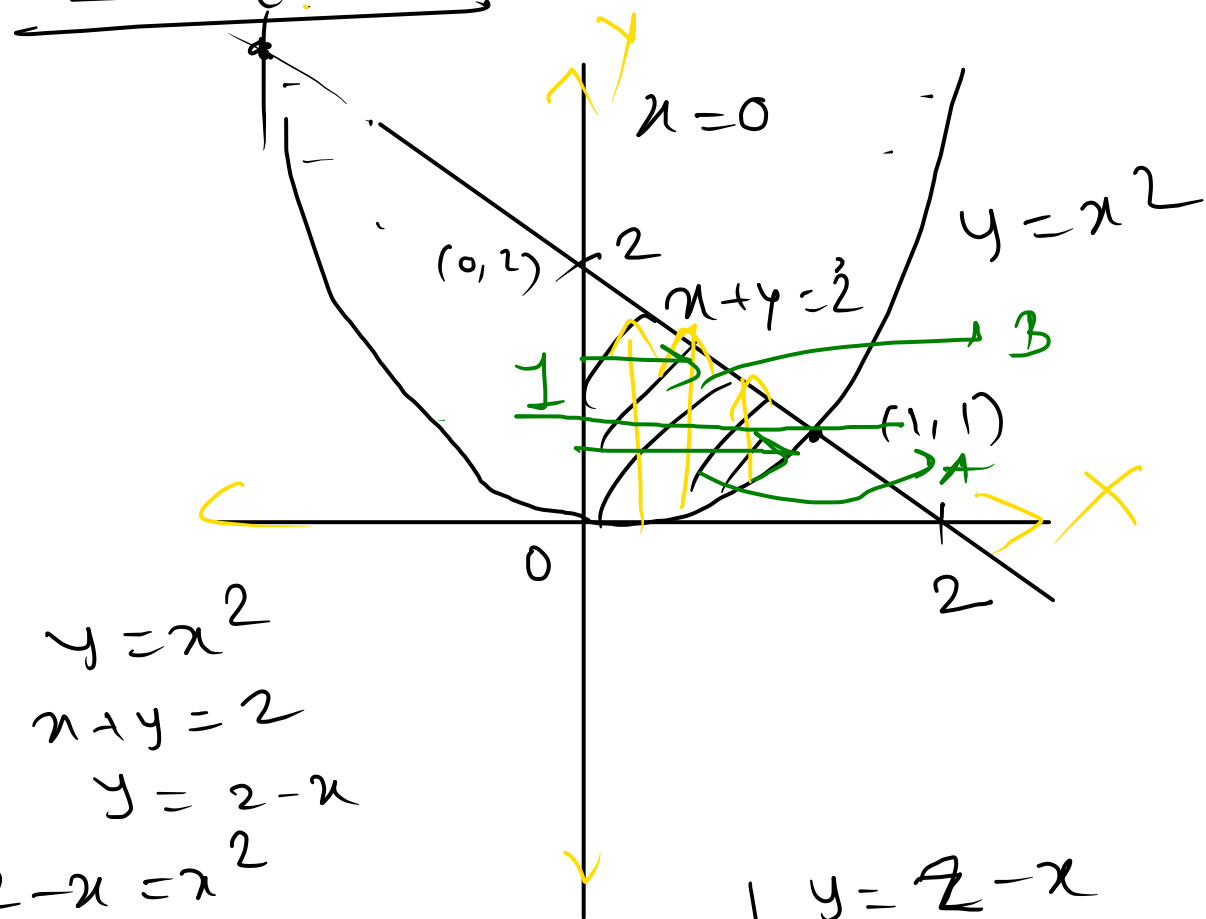
$x+y=2$ in 1st quadrant

$$y=x^2$$

$$x = \pm \sqrt{y}$$

$$x+y=2$$

x	0	2	
y	2	0	0



$$y=x^2$$

$$x+y=2$$

$$y=2-x$$

$$2-x=x^2$$

$$\Rightarrow x^2+x-2=0$$

$$x=1, -2$$

$$y=2-x$$

$$x=1, -2$$

$$y=1, 4$$

$$\int_{y=0}^1 \int_{x=0}^{\sqrt{y}} y \, dx \, dy$$

$$+ \int_{y=1}^2 \int_{x=0}^{2-y} y \, dx \, dy$$

$$\int_{x=0}^1 \int_{y=x^2}^{2-x} y \, dy \, dx = \frac{16}{15}$$

$$\iint_R \frac{y}{(a-x)\sqrt{ax-y^2}} dx dy$$

R is region bdd by $y^2=ax$ & $y=x$

$$y = \pm \sqrt{ax}$$

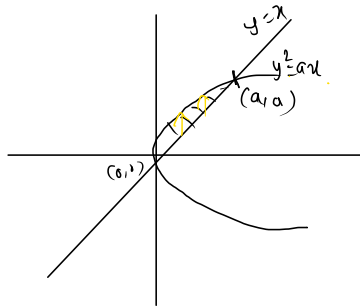
$$y=x \quad y^2=ax$$

$$x^2=ax$$

$$x(x-a)=0$$

$$x=0, x=a$$

$$y=x \quad \begin{matrix} x & 0 & a \\ y & 0 & a \end{matrix}$$



$$\int_{x=0}^a \int_{y=x}^{\sqrt{ax}} \frac{y}{(a-x)\sqrt{ax-y^2}} dy dx$$

$$= \int_0^a \frac{1}{(a-x)} \int_x^{\sqrt{ax}} \frac{y}{\sqrt{ax-y^2}} dy dx$$

$$0 - 2y dy = dt \Rightarrow y dy = -\frac{dt}{2}$$

$$\text{as } y: x \rightarrow \sqrt{ax} \quad t: ax-x^2 \rightarrow 0$$

$$= \int_0^a \frac{1}{a-x} \int_{ax-x^2}^0 \frac{-\frac{dt}{2}}{2\sqrt{t}} dx$$

$$= \frac{1}{2} \int_0^a \frac{1}{a-x} \int_0^{ax-x^2} t^{-1/2} dt dx$$

$$= \frac{1}{2} \int_0^a \frac{1}{a-x} \left[\frac{t^{1/2}}{1/2} \right]_0^{ax-x^2} dx$$

$$= \frac{1}{2} \int_0^a \frac{1}{a-x} [2(ax-x^2)^{1/2} - 0] dx$$

$$= \frac{2}{2} \int_0^a \frac{1}{a-x} x^{1/2} (a-x)^{1/2} dx \quad \left[\int_0^1 x^m (1-x)^n dx = \frac{\beta(m+1, n+1)}{\beta(m+1, n+1)} \right]$$

$$= \int_0^a x^{1/2} (a-x)^{-1/2} dx$$

$$= a^{-1/2} \int_0^a x^{1/2} \left(1 - \frac{x}{a}\right)^{-1/2} dx$$

$$\frac{x}{a} = t \Rightarrow x = at$$

$$dx = a dt$$

$$= a^{-1/2} \int_0^1 (at)^{1/2} (1-t)^{-1/2} a dt$$

$$= a^{-1/2} a^{1/2} a \int_0^1 t^{1/2} (1-t)^{-1/2} dt = a \beta\left(\frac{3}{2}, \frac{1}{2}\right)$$

$$= a \frac{\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma 2}$$

$$\Gamma(n) = (n-1) \Gamma(n-1) //$$

$$= \frac{a}{1} \frac{1}{2} \Gamma\left(\frac{3}{2}\right) \sqrt{\pi}$$

$$= \frac{\pi a}{2}$$

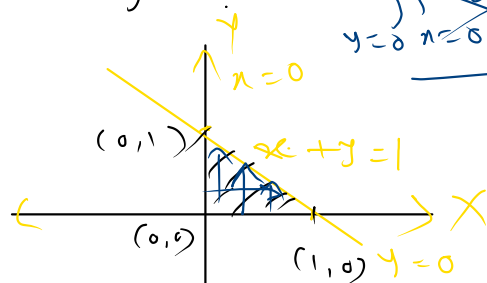
$\iint \sqrt{xy(1-x-y)} \, dx \, dy$ over the area

bdd by $x=0$, $y=0$ & $x+y=1$.

~~$\int_0^1 \int_0^{1-y} dx \, dy$~~
 $y=0 \quad x=0$

$x+y=1$

$x \quad y \quad 1$
 $0 \quad 1 \quad 0$



$\int_0^1 x^m (1-x)^n dx$
 $=$

$x=1 \quad y=1-x$
 $x=0 \quad y=0$
 $\iint \sqrt{xy(1-x-y)} \, dy \, dx$

$= \int_0^1 \sqrt{x} \int_0^{1-x} \sqrt{y} \sqrt{1-x-y} \, dy \, dx$

$= \int_0^1 \sqrt{x} \int_0^{1-x} y^{1/2} (1-x-y)^{1/2} \, dy \, dx$

$= \int_0^1 \sqrt{x} \int_0^a a^{1/2} y^{1/2} (1-y/a)^{1/2} \, dy \, dx$

put $y/a = t \Rightarrow y = at$
 $dy = a \, dt$
 $y: 0 \rightarrow a \quad t: 0 \rightarrow 1$

$= \int_0^1 \sqrt{x} a^{1/2} \int_0^1 (at)^{1/2} (1-t)^{1/2} a \, dt \, dx$

$= \int_0^1 \sqrt{x} a^{1/2} a^{1/2} a \int_0^1 t^{1/2} (1-t)^{1/2} \, dt \, dx$

$= \int_0^1 \sqrt{x} a^2 \, \beta(3/2, 3/2) \, dx = \frac{\sqrt{3/2} \sqrt{3/2}}{\sqrt{3}} \int_0^1 x^{1/2} (1-x)^{1/2} \, dx$

$= \frac{(\sqrt{3/2})^2}{2} \beta(3/2, 3/2)$

$= \frac{\pi}{4\pi^2} \frac{\sqrt{3/2} \sqrt{3}}{\sqrt{9/2}}$

$= \frac{\pi}{8} \frac{\sqrt{3/2} \cdot 2}{\pi \cdot \sqrt{3/2} \cdot \sqrt{3/2}}$

$= \frac{2\pi}{105}$

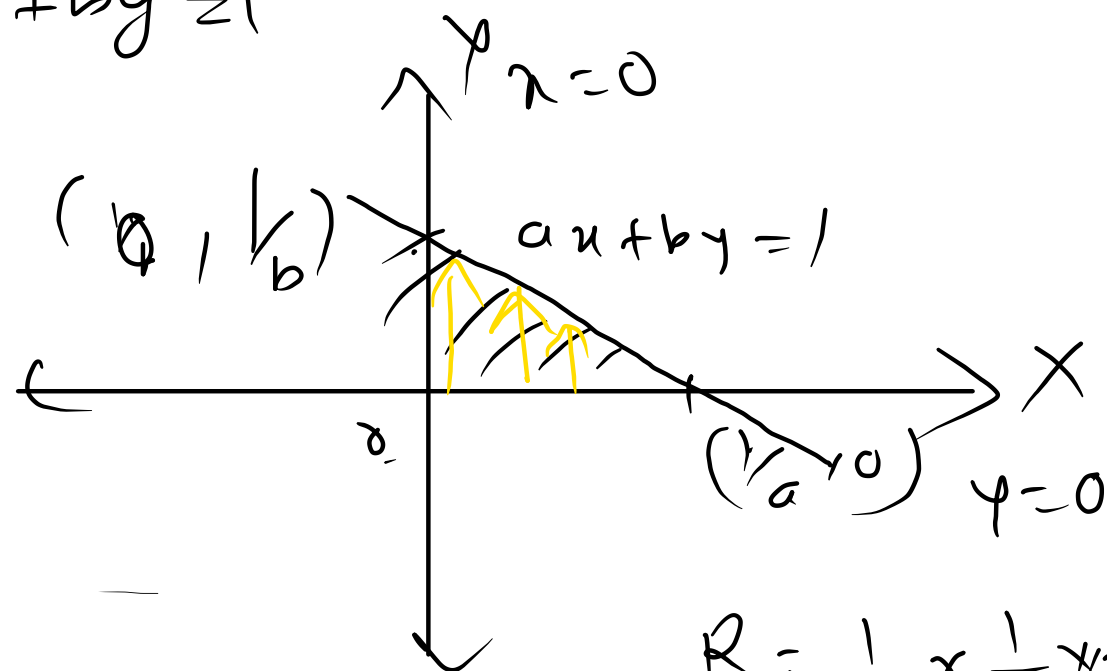
HW:

P.T. $\iint_R e^{ax+by} dx dy = 2R$ where R is

area of Δ whose boundaries are

$x=0, y=0$ & $ax+by=1$

$ax+by=1$



$$\int_{x=0}^{1/a} \int_{y=0}^{1/b(1-ax)} e^{ax+by} dy dx = \frac{1}{ab}$$

$= \frac{1}{ab}$

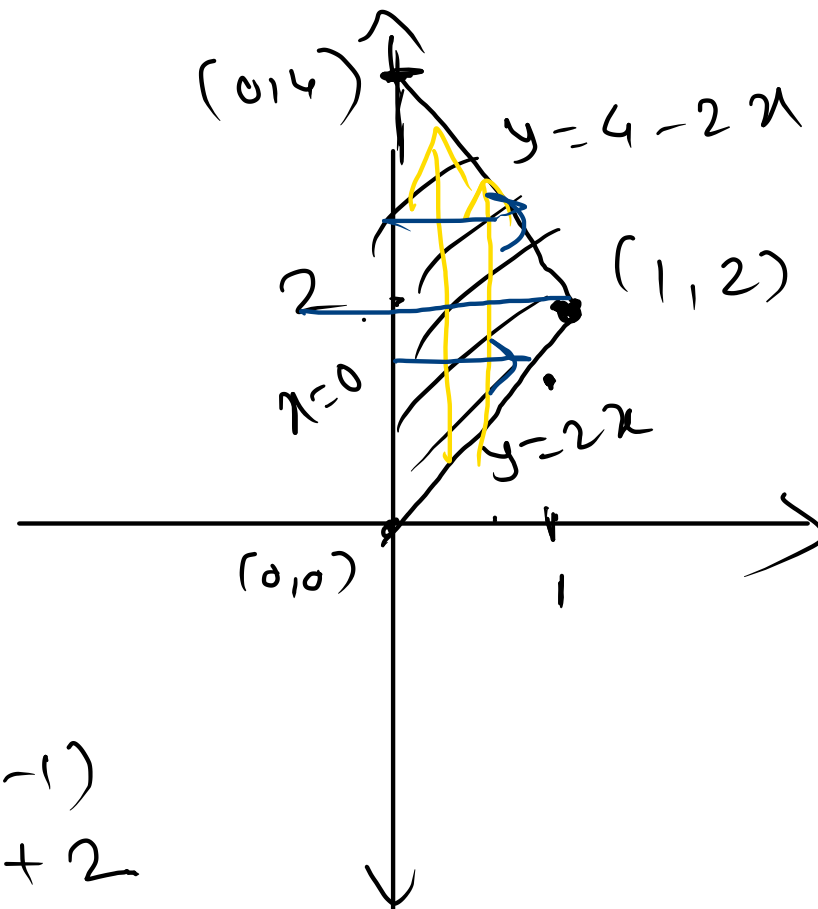
$R = \frac{1}{2} \times \frac{1}{a} \times \frac{1}{b}$

$\iint_R x(x-y) \, dx \, dy$ where R is Δ with vertices $(0,0)$ $(1,2)$ $(0,4)$.

$$\frac{(y-y_1)}{(x-x_1)} = \frac{y_2-y_1}{x_2-x_1}$$

$$\begin{array}{cc} (0,0) & (1,2) \\ \frac{y}{x} = \frac{2}{1} & \Rightarrow y = 2x \end{array}$$

$$\begin{array}{cc} (1,2) & (0,4) \\ \frac{y-2}{x-1} = \frac{2}{-1} & \Rightarrow y-2 = -2(x-1) \\ & \Rightarrow y-2 = -2x+2 \\ & \Rightarrow y = 4-2x \end{array}$$



✓

$$\int_{x=0}^1 \int_{y=2x}^{4-2x} x(x-y) \, dy \, dx$$

$x=0$ $y=2x$

$=$

$$\int_{y=0}^2 \int_{x=0}^{y/2} x(x-y) \, dx \, dy$$

$+ \int_{y=2}^4 \int_{x=0}^{4-y/2} x(x-y) \, dx \, dy$

✗

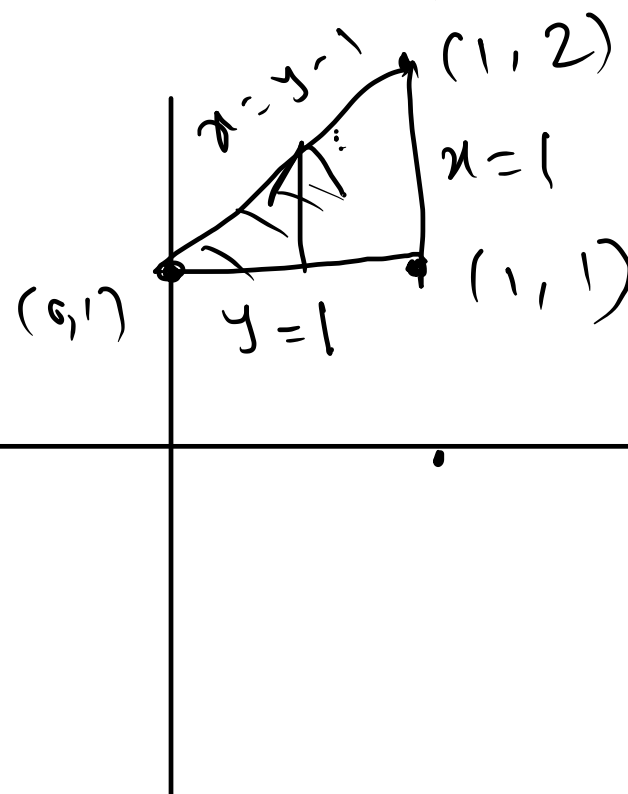
(HW)

$\iint (x^2 + y^2) dx dy$ over area of Δ

whose vertices are $(0,1)$ $(1,1)$ $(1,2)$

$$\int_{x=0}^1 \int_{y=1}^{1+x} (x^2 + y^2) dy dx = \frac{7}{6}$$

$$\frac{y-1}{x} = \frac{1}{1}$$

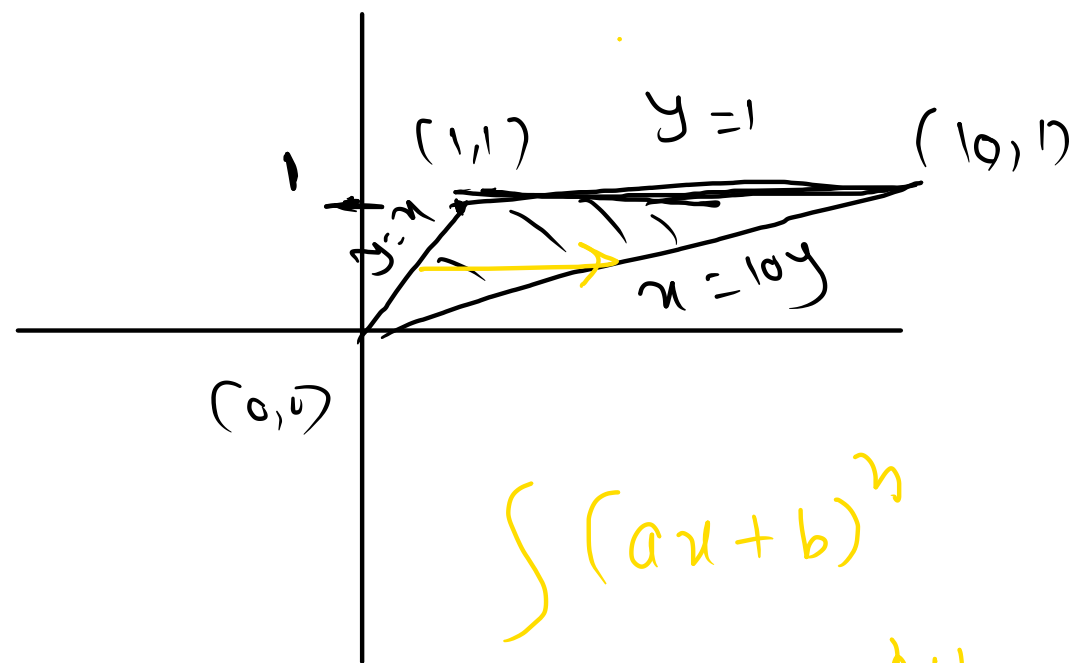


(HW)

$$\iint_R \sqrt{xy - y^2} \, dx \, dy \quad \text{where } R \text{ is } \Delta$$

vertices are $(0,0)$ $(10,1)$ $(1,1)$

$$\begin{matrix} (0,0) & (10,1) \\ \frac{y}{x} = \frac{1}{10} \Rightarrow 10y = x \end{matrix}$$



$$\int_{y=0}^1 \int_{x=y}^{x=10y} \sqrt{xy - y^2} \, dx \, dy$$

$$= \int_0^1 \int_y^{10y} y^{1/2} (x-y)^{1/2} \, dx \, dy$$

$$= \int_0^1 y^{1/2} \left[\frac{(x-y)^{3/2}}{3/2} \right]_y^{10y} \, dy$$

$$\begin{aligned} \int (ax+b)^n &= \frac{(ax+b)^{n+1}}{(n+1)a} \end{aligned}$$

$$= 6$$

$\int \int x^{m-1} y^{n-1} dx dy$ over the
quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$\int_{x=0}^a \int_{y=0}^{b\sqrt{1-\frac{x^2}{a^2}}} x^{m-1} y^{n-1} dy dx$$

$$= \int_0^a x^{m-1} \left[\frac{y^n}{n} \right]_0^{b\sqrt{1-\frac{x^2}{a^2}}} dx$$

$$= \frac{b^n}{n} \int_0^a x^{m-1} \left(1 - \frac{x^2}{a^2}\right)^{n/2} dx$$

$$= \frac{b^n}{n} \int_0^a x^{m-1} \left(1 - \frac{x^2}{a^2}\right)^{n/2} dx$$

$$\frac{x^2}{a^2} = t$$

$$x: 0 \rightarrow a$$

$$t: 0 \rightarrow 1$$

$$\Rightarrow x^2 = a^2 t$$

$$x = a t^{1/2}$$

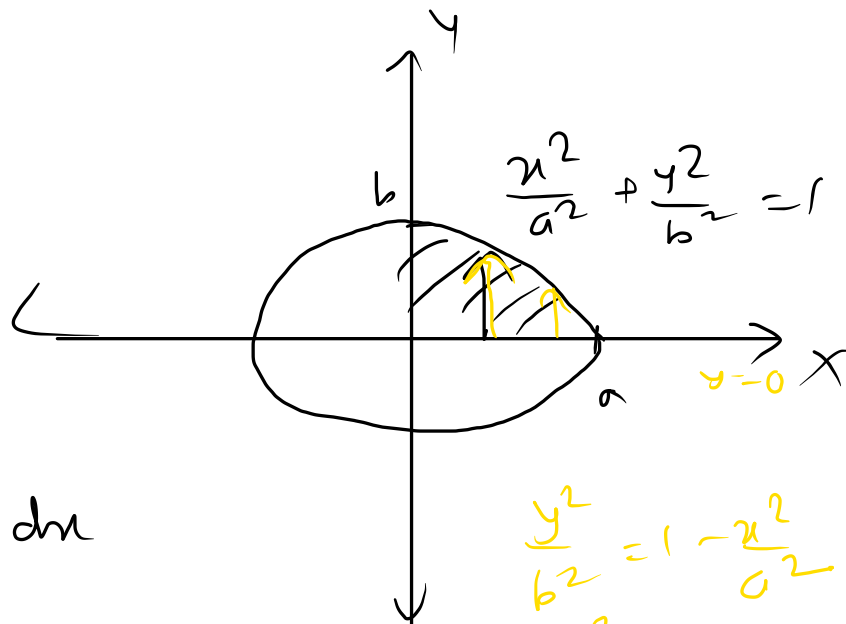
$$dx = a \frac{1}{2} t^{-1/2} dt$$

$$\left(b \sqrt{1-\frac{x^2}{a^2}} \right)^n$$

$$= \frac{b^n}{n} \int_0^1 (a t^{1/2})^{m-1} (1-t)^{n/2} a \frac{1}{2} t^{-1/2} dt$$

$$= \frac{b^n}{n} \frac{a^{m-1} a}{2} \int_0^1 t^{\frac{m}{2}-\frac{1}{2}-\frac{1}{2}} (1-t)^{n/2} dt$$

$$= \frac{b^n a^m}{2n} \int_0^1 t^{\frac{m}{2}-1} (1-t)^{n/2} dt = \frac{b^n a^m}{2n} \beta\left(\frac{m}{2}, \frac{n}{2}+1\right)$$



$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = b^2 \left(1 - \frac{x^2}{a^2}\right)$$

$\iint_R xy \, dx \, dy$ over the region R given

by $\underline{x^2 + y^2 - 2x = 0}$, $\underline{y^2 = 2x}$, $\underline{y = x}$

$$x^2 - 2x + \frac{4}{4} + y^2 = 1$$

$$(x-1)^2 + y^2 = 1$$

(center $(1, 0)$ & rad 1)

$$y = \pm \sqrt{2x}$$

$$x=1 \quad y = \pm \sqrt{2}$$

$$y^2 = 2x$$

$$x^2 = 2x$$

$$y = x$$

$$\Rightarrow x(x-2) = 0$$

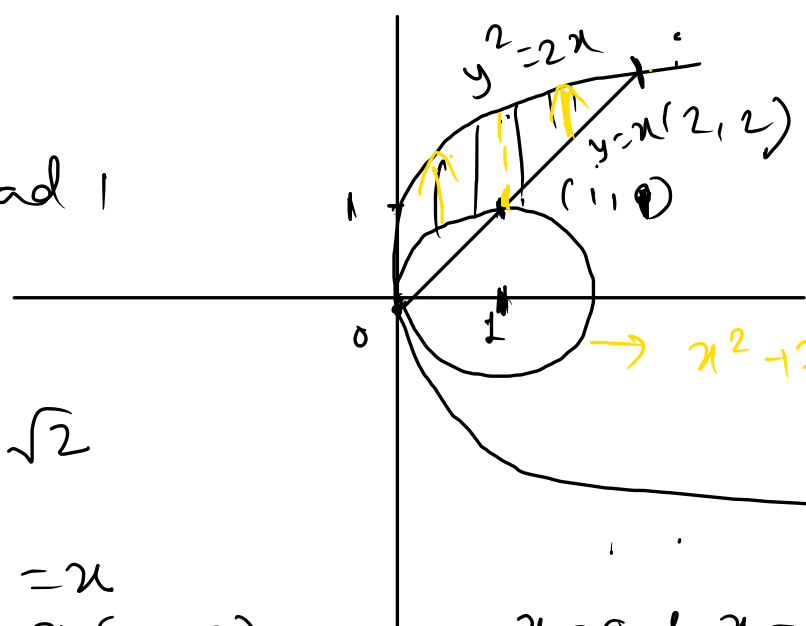
$$\Rightarrow x=0 \text{ & } x=2$$

$$y=0$$

$$y=2 \quad (y=x)$$

$$\int_{x=0}^1 \int_{y=\sqrt{2x-x^2}}^{\sqrt{2x}} xy \, dy \, dx + \int_{x=1}^2 \int_{y=x}^{\sqrt{2x}} xy \, dy \, dx$$

$$= 7/12$$



$$\rightarrow x^2 + y^2 - 2x = 0$$

$$y^2 = 2x - x^2$$

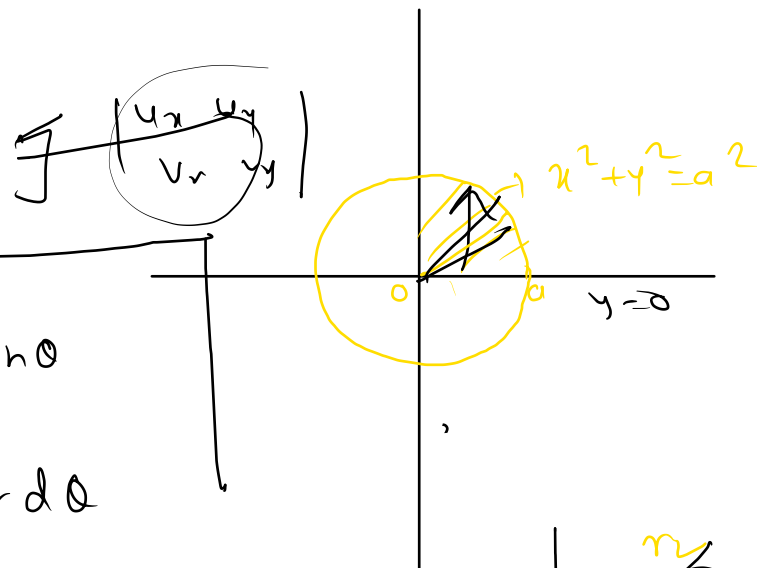
$$y = \pm \sqrt{2x - x^2}$$

$$\frac{m^2}{4f}$$

$\iint (x^2 - y^2) x \, dx \, dy$ over the +ve quadrant of circle $x^2 + y^2 = a^2$

$$\int_0^a \int_0^{\sqrt{a^2 - x^2}} (x^2 - y^2) x \, dy \, dx$$

$x \geq 0, y \geq 0$



Polar Co-ordinates

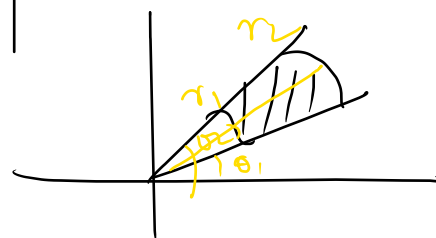
$$x = r \cos \theta, \quad y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$dy \, dx = r \, dr \, d\theta$$

$$\theta: \theta_1 \rightarrow \theta_2$$

$$r: r_1 \rightarrow r_2$$



$$\iint (x^2 - y^2) x \, dx \, dy$$

$$\int_0^{\pi/2} \int_0^a (r^2 \cos^2 \theta - r^2 \sin^2 \theta) r \cos \theta \, r \, dr \, d\theta$$

$\theta \geq 0, r \geq 0$

$$= \int_0^{\pi/2} \int_0^a r^2 (\cos 2\theta) r^2 \cos \theta \, dr \, d\theta$$

$$= \int_0^{\pi/2} \cos 2\theta \cos \theta \left(\frac{r^5}{5} \right)_0^a d\theta = \frac{a^5}{5} \int_0^{\pi/2} \frac{\cos(3\theta) + \cos \theta}{2} d\theta$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$= \frac{a^5}{5 \times 2} \left[\frac{\sin 3\theta}{3} + \sin \theta \right]_0^{\pi/2}$$

$$= \frac{a^5}{10} \left[-\frac{1}{3} + 1 \right] = \frac{a^5}{10} \times \frac{2}{3}$$

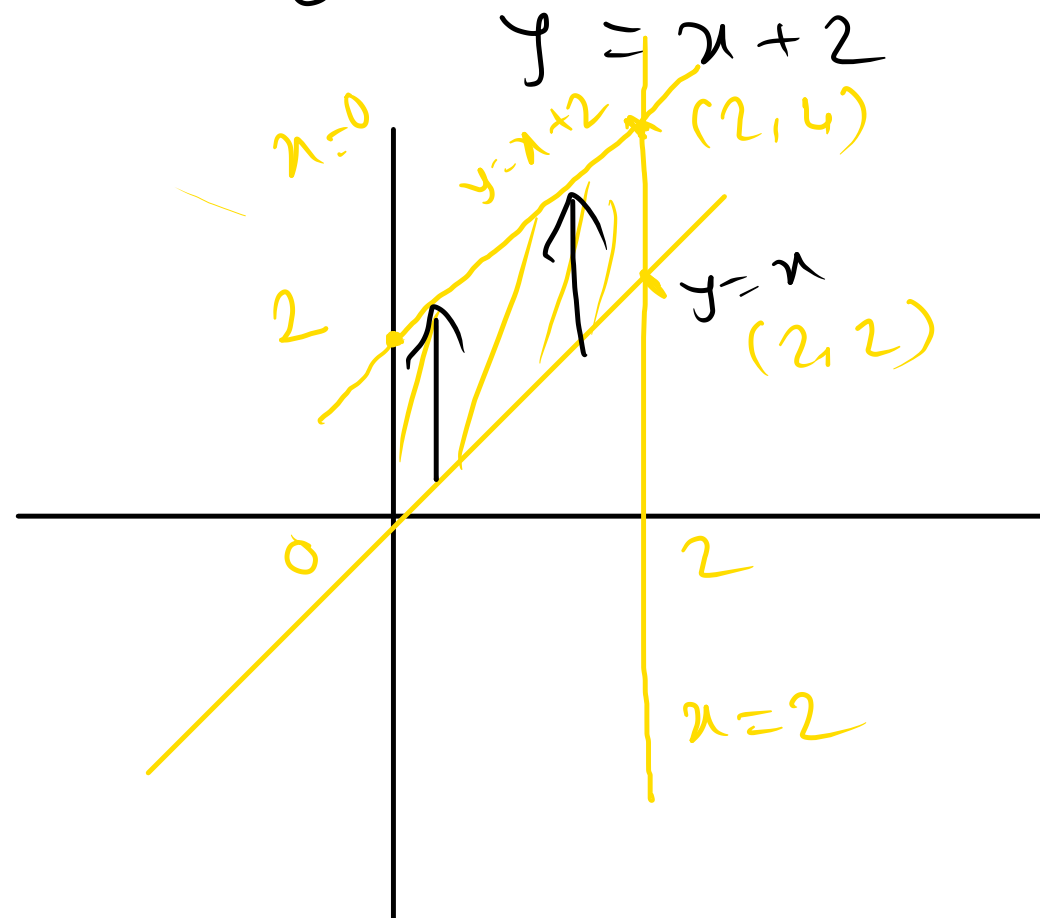
evaluate $\iint_R (x+y) dx dy$ where R

is the region bdd by $x=0$, $x=2$, $y=x$

$$y = x + 2$$

$$x \quad 0 \quad 2$$

$$y \quad 2 \quad 4$$



$$\int_0^2 \int_{x+2}^{x+4} (x+y) dy dx$$

$$= 12$$

$\iint_R x^2 dA$ where R is region in 1st quad.

bdd by hyperbola $xy=16$ & line
 $x=y$, $y=0$ & $x=8$.



$$xy = 16$$

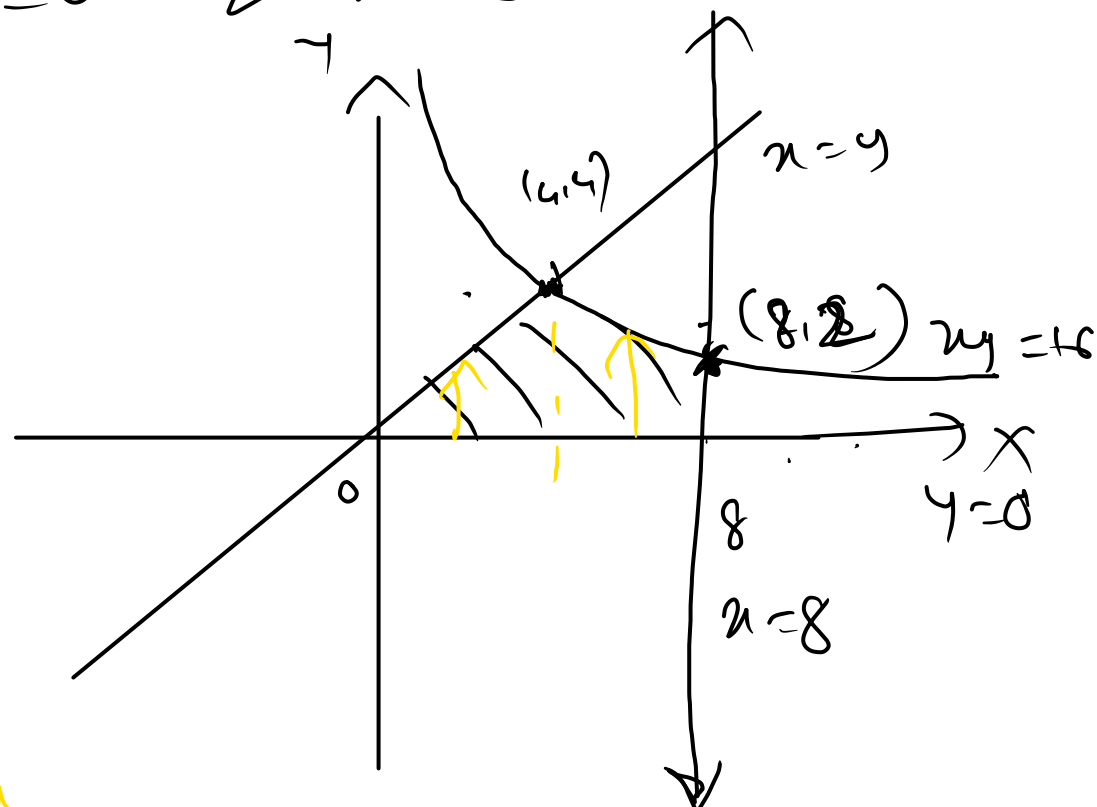
$$x=8 \quad y=2$$

$$xy = 16 \quad x=y$$

$$y^2 = 16$$

$$y=4$$

$$x=4$$



$$\iint_R x^2 dA = \int_{x=0}^4 \int_{y=0}^x x^2 dy dx + \int_{x=4}^8 \int_{y=0}^{16/x} x^2 dy dx = 448$$